LING285-L5&6

Course: LING285: Human Language and Technology

Title of Lecture: Lecture 5 & 6: Dynamical Systems 3 & 4

Date: 9/8&11/2026

Professor: Khalil Iskarous

Section 1: The Elements of the Harmonic Oscillator, x and v

- A **harmonic oscillator** is built from only two quantities, x and v. We can think of x as how far something is displaced from its resting point, and v as its **velocity**, which is how fast and in which direction x is moving. Neither variable is interesting alone. The whole system is about **how x and v change over time as a result of their relation to each other**.
- To talk about change over time, we put a dot over a variable. $\dot{x}$ means the change in x as time increases, and $\dot{v}$ means the change in v as time increases. This way of writing rates of change is **Isaac Newton's notation**.
    - The sign of the dotted quantity tells us the direction of change. When $\dot{x}$ is positive, x is rising, and when $\dot{x}$ is negative, x is falling.
    - The same rule holds for v. A positive $\dot{v}$ means v is rising, and a negative $\dot{v}$ means v is falling.
    - The size of the dotted quantity tells us how quickly the change happens. A large positive $\dot{x}$ means x is climbing quickly, and a small positive $\dot{x}$ means it is climbing slowly.
- The defining feature of the harmonic oscillator is that x **and** v **change each other**. The rate at which x changes is set by v, and the rate at which v changes is set by x. Neither variable can change on its own terms, and this two-way dependence is what produces repeating back-and-forth motion.

Section 2: The Two Equations of the Harmonic Oscillator

- The first equation says that the rate of change of x equals the current value of v:

$$\dot{x} = v$$

- When v is positive, $\dot{x}$ is positive, so x rises.
- When v is negative, $\dot{x}$ is negative, so x falls.
- In other words, v directly steers x. Whatever sign v has at a given moment is the direction x moves at that moment.
- The second equation says that the rate of change of v equals the negative of x:

$$\dot{v} = -x$$

- The **minus sign** is the heart of the system. It means v is always pushed in the direction opposite to wherever x currently is.
- When x is positive, $\dot{v}$ is negative, so v falls. As v keeps falling, it eventually becomes negative, and through the first equation that negative v makes x fall back down.
- When x is negative, $\dot{v}$ is positive, so v rises. Once v becomes positive, the first equation makes x rise back up.
- Putting the two equations together, whenever x moves away from zero in either direction, the second equation turns v around, and the first equation then carries x back toward zero. This **restoring relationship** is what keeps the system from drifting off and instead makes it swing back and forth.
- The relationship can be drawn as a **loop of influence** with x and v each in a circle. One arrow runs from v to x, standing for $\dot{x} = v$, since v determines how x changes. A second arrow runs from x back to v, standing for $\dot{v} = -x$, since x determines how v changes.

The loop is closed, so a change in v changes x, and that change in x goes on to change v again.

Section 3: A Worked Example with x and v

- Suppose that at some moment v equals -5. Substituting into the first equation gives:

$$\dot{x} = v = -5$$

- The rate of change of x is negative, so x **will go down**.
- The value of x itself does not matter for this conclusion. Only the sign of v decides whether x is rising or falling.
- Now suppose that at some moment x equals -5. Substituting into the second equation gives:

$$\dot{v} = -x = -(-5) = 5$$

- The two minus signs cancel, so the rate of change of v is positive, and v **will go UP**.
- This is the restoring behavior in action. When x is below zero, v is pushed upward, and an upward v will then lift x back toward zero through the first equation.
- The example shows why x and v must be read together. Knowing v tells us what x is about to do, and knowing x tells us what v is about to do.

Section 4: The x-v System Traced Through Time

- We can follow the system moment by moment, starting with x at its highest positive value and v equal to zero.
    - At the start, x is positive, so $\dot{v}$ is negative and v begins to drop below zero. As soon as v is negative, $\dot{x}$ is negative, so x starts to decrease. At first x decreases slowly, since v is only slightly negative.
    - While x is still positive, v keeps getting more negative. A more negative v makes x fall faster, so x drops more and more steeply.
    - When x reaches zero, $\dot{v}$ equals zero, so v stops decreasing. At this instant v is at its most negative value, which means x is crossing zero at its fastest speed.
    - Once x is below zero, $\dot{v}$ becomes positive, so v starts rising back toward zero. x is still falling, but more and more slowly.
    - When v returns to zero, x has reached its lowest, most negative value. At that point x stops moving down and turns around.
    - With x negative, v keeps rising and becomes positive, so x climbs. v reaches its largest positive value just as x crosses zero on the way up, and then x returns to its maximum as v comes back to zero. The system is now back where it started, and the whole cycle repeats.
- Computing the system at a series of separate moments in time gives a sequence of points, where each new value of x and v is worked out from the values just before it. Connecting these points produces two smooth waves.
    - The x wave starts at the top, dips below zero to a lowest point, and climbs back up past zero to the top again.
    - The v wave starts at zero, dips below zero to its lowest point, returns to zero, and then rises above zero.
    - The two waves have the same shape but are shifted in time by a quarter of a cycle. v is at its extreme values exactly when x crosses zero, and v is zero exactly when x is at its extreme values.

Section 5: The Oscillation/Vibration Dynamical System

- A **dynamical system** is a set of variables together with rules for how those variables change over time. The x-v system is a specific kind called an **oscillation** or **vibration dynamical system**. Oscillation and vibration mean the same thing here: a quantity repeatedly swinging back and forth around a middle value.

- The loop between x and v is enough, by itself, to generate this repeated swinging. x drives changes in v, and v drives changes in x, so once the system is set in motion it keeps cycling without anything outside it having to push it again.
- Plotted against time t, x traces a wave that starts at its peak, falls below zero, rises back to the same peak, and repeats. Each cycle reaches the same height and the same depth, and each cycle takes the same amount of time. v follows the same kind of repeating wave on its own axis over the same stretch of time.

Section 6: Everything's a Harmonic Oscillator, If You're Brave Enough

- The same two-variable structure of $\dot{x} = v$ and $\dot{v} = -x$ shows up in a huge range of systems, including the air in the vocal tract, the vocal folds, economies, music, and matter itself.
- Calling a system a harmonic oscillator means identifying something in it that plays the role of x, a displacement away from a resting state, and something that plays the role of v, the rate at which that displacement changes. It also means accepting a simplification, since real systems are usually messier than two perfectly linked variables. The harmonic oscillator is still useful as a model, since it captures the basic back-and-forth pattern that all of these systems share.

Section 7: A Slice of Air in the Vocal Tract

- The **vocal tract** is the air-filled passage from the vocal folds in the throat up through the mouth to the lips. We can picture a thin slice of air inside it, containing a large number of air particles.
- **Pressure** in this slice depends on how close together the particles are.
  - **High pressure** means the air particles are very close together.
  - **Low pressure** means the air particles are very far apart.
  - The particles themselves stay the same. There are the same number of them, and each one is the same size. The only thing that changes is the distance between them.
- **Compression** is the state in which particles are squeezed closer together than their normal spacing. **Rarefaction** is the state in which particles are spread farther apart than their normal spacing. The normal spacing, where the air is neither squeezed nor stretched, is the **equilibrium spacing**.
  - If we track one pair of neighboring particles, at equilibrium they sit at 100% of their equilibrium distance.
  - At the strongest compression, that pair is squeezed to about 56% of its equilibrium distance, so the particles are a little more than half as far apart as usual.
  - At the strongest rarefaction, the same pair is stretched to about 141% of its equilibrium distance.
  - The slice cycles continuously between these states. It compresses, passes back through equilibrium, rarefies, and passes back through equilibrium again. In real speech the motion is much smaller than these exaggerated values, and the slice is an idealized piece of the air.
- The slice of air behaves like a harmonic oscillator. The spacing of the particles, measured relative to equilibrium, plays the role of x. Air that is squeezed together tends to push itself apart, and air that is spread out tends to be pulled back together, so any departure from equilibrium spacing is pushed back toward equilibrium. This is the same restoring relationship expressed by $\dot{v} = -x$, and it is why the slice keeps swinging between compression and rarefaction.

Section 8: There's a Cosine in Your Mouth

- The back-and-forth motion of the air slice follows a **cosine** function:

$$x = \cos(\text{phase})$$

- Here x measures the **normalized expansion** of the slice. Normalized means the values are scaled so that the largest stretch equals $+1$ and the largest squeeze equals -1.

    - $x = +1$ means the particles are **farthest apart**, which is maximum rarefaction.
    - $x = 0$ means the particles are at **equilibrium** spacing.
    - $x = -1$ means the particles are **closest together**, which is maximum compression.
    - Any positive value of x means the slice is rarefied to some degree, and any negative value means it is compressed to some degree.
- The **phase** is the position within a cycle. One complete cycle corresponds to the phase increasing by 2π. Running the phase from 0 to 6π therefore covers exactly three full cycles, with the first cycle ending at 2π, the second at 4π, and the third at 6π.
- Following one cycle in order:
    - At phase 0, the cosine equals $+1$, so the slice is at maximum rarefaction.
    - As the phase increases, x decreases. A value such as $x = +0.74$ means the particles are still farther apart than equilibrium, but less so than at the start.
    - At phase $\pi/2$, the cosine equals 0, so the slice passes through equilibrium.
    - A value such as $x = -0.54$ means the slice is now compressed, with particles closer together than equilibrium.
    - At phase π, the cosine equals -1, so the slice reaches maximum compression.
    - At phase $3\pi/2$, x returns to 0, and at phase 2π, x is back at $+1$, completing the pattern of maximum rarefaction, then compression, then rarefaction again.
    - Over the next two cycles, x passes through values such as $+0.31$, -0.05, -0.21, and $+0.45$, and each value's sign tells us whether the slice is rarefied or compressed at that moment.
- The cosine is the same wave that the x-v system produces when x starts at its maximum and v starts at zero. In both cases the curve begins at the top and swings downward. So the vibrating air in the vocal tract is following the harmonic oscillator relationship, and its expansion over time traces a cosine.

[CONTINUE]

Section 9: Vowels and the Shape of the Vocal Tract

- Vowels differ from one another in where the **tongue** sits inside the vocal tract. These positions can be imaged with magnetic resonance imaging, which produces a picture of the head cut down the middle from front to back, showing the lips, tongue, roof of the mouth, and throat. In these images the speaker faces left, the tongue is the large bright mass in the mouth, and the dark regions are air.
- Vowels are organized along two dimensions. **Height** describes whether the tongue body is raised toward the roof of the mouth (**high**) or lowered (**low**). **Frontness** describes whether the tongue body is pushed forward toward the teeth (**front**), pulled back toward the throat (**back**), or held in between (**central**).
    - In **"dean,"** the vowel is **high and front**. The tongue body is raised and pushed forward, close to the hard roof of the mouth, which leaves a large open air space in the throat behind the tongue.
    - In **"dune,"** the vowel is **high and back**. The tongue is still raised, but its highest point is pulled back toward the soft rear part of the roof of the mouth.
    - In **"Dan,"** the vowel is **low and front**. The tongue is lowered and the mouth is more open, with the tongue body still positioned toward the front.
    - In **"Don,"** the vowel is **low and back**. The tongue is lowered and pulled back into the throat, which narrows the air space behind it.
    - In **"done,"** the vowel is **central**. The tongue rests in the middle of the mouth, neither strongly raised nor lowered and neither pushed forward nor pulled back.
- Each tongue position gives the air in the vocal tract a different shape. A differently shaped column of air vibrates in a different way, and that difference in vibration is what makes one vowel sound different from another.

Section 10: Vowels as Combined Vibrations

- A single cosine describes air vibrating at one frequency. A vowel is more complex than that. The air in the vocal tract vibrates at **several frequencies at the same time**, and the vowel is the **sum** of those vibrations. Since each individual vibration is a cosine, and each cosine is the behavior of a harmonic oscillator, a vowel can be modeled as **harmonic oscillators added together**.
- **Frequency** is the number of complete cycles a vibration makes per second, measured in **hertz (Hz)**. A vibration at f hertz can be written as $\cos(2\pi ft)$, where t is time in seconds. Multiplying by 2π converts the number of cycles into phase, since one full cycle equals a phase of 2π.
- The vowel /**i**/, the vowel in "dean," is modeled as a combination of a 1 Hz vibration and a 4 Hz vibration:

$$x(t) = \cos(2\pi t) + \cos(8\pi t)$$

- The first term is the 1 Hz vibration, since $2\pi \times 1 = 2\pi$. The second term is the 4 Hz vibration, since $2\pi \times 4 = 8\pi$.
- The result is a slow wave with fast ripples riding on top of it. Over 3 seconds, the slow component completes 3 cycles while the fast component completes 12 cycles.
- The vowel /**a**/, the vowel in "Don," is modeled as a combination of a 2 Hz vibration and a 3 Hz vibration:

$$x(t) = \cos(4\pi t) + \cos(6\pi t)$$

- The first term is the 2 Hz vibration, since $2\pi \times 2 = 4\pi$. The second term is the 3 Hz vibration, since $2\pi \times 3 = 6\pi$.
- Over 3 seconds, the 2 Hz component completes 6 cycles and the 3 Hz component completes 9 cycles. The two frequencies are close together, so the sum produces a different repeating pattern of tall and short swings than /i/ does.
- These frequencies are **illustrative**. They are chosen low enough that the motion can be watched, and the two components in each vowel are given **equal amplitudes**, meaning each cosine swings between $+1$ and -1.
  - Since each component reaches at most $+1$ and at least -1, their sum ranges between $+2$ and -2.
  - The sum reaches $+2$ only when both cosines peak at the same moment. For both vowels this happens at $t = 0, 1, 2$, and 3 seconds, since every component frequency completes a whole number of cycles in each second.
  - At other moments the components partly cancel or partly reinforce each other. For example, /i/ can take a value like -1.21 at one moment while /a/ takes a value like $+0.58$, and at another moment /i/ can be at $+0.62$ while /a/ is at -1.62.
- The same sign rule from the single cosine still applies. **Positive x means rarefaction**, with particles farther apart than equilibrium, and **negative x means compression**, with particles closer together. For /i/ and /a/, the slice of air in the vocal tract is squeezed and stretched in two different time patterns, and those two patterns are what distinguish the two vowels. Watching these patterns at one-quarter speed makes the different rhythms of squeezing and stretching visible.

Section 11: Praat Demonstration

- **Praat** is a free computer program used in phonetics to record and analyze speech. It displays the **waveform** of a recorded sound, which is the air pressure changing over time, and it can also break that sound into the frequencies it contains.
- Recording vowels such as those in "dean" and "Don" in Praat shows the combined vibrations directly. Each vowel produces its own repeating waveform, and each waveform is made of several frequencies added together, which is the same idea as the two-frequency models of /i/ and /a/.

Section 12: Two Things Are Vibrating, the Vocal Tract Air and the Vocal Folds

- During speech, **two things vibrate**. The first is the air in the vocal tract, which is the vibration modeled by the cosines and sums of cosines above. The second is the **vocal**

folds.

- The vocal folds are two folds of tissue in the **larynx**, the voice box in the throat. When air from the lungs pushes up against the closed folds, pressure builds until the folds are forced apart. Once they open and air rushes through, the pressure between them drops and the folds come back together. The cycle then repeats many times per second.
    - This repeated opening and closing is itself an oscillation, with the folds swinging away from their closed position and being pulled back again.
    - Each opening lets a puff of air through, so the vocal folds chop the steady airflow from the lungs into a rapid series of puffs.
    - Those puffs set the air in the vocal tract vibrating, and the shape of the vocal tract, set by the tongue position, determines which combination of frequencies that air vibrates at.
- Speech sounds such as vowels therefore come from one oscillator driving another. The vocal folds vibrate, and the air in the vocal tract vibrates in response.

Section 13: The Harmonic Oscillator in Economics

- The harmonic oscillator pattern also appears in the **business cycle**, the repeated rise and fall of economic activity over time. The business cycle is plotted with **time** on the horizontal axis and the **level of real output** on the vertical axis. **Real output** is the total amount of goods and services an economy produces, adjusted so that price changes do not distort the comparison across years.
- The business cycle moves through a repeating sequence of phases.
    - At a **peak**, output is at a high point.
    - During a **recession**, output falls.
    - At a **trough**, output reaches a low point.
    - During a **recovery**, output rises again until it reaches the next peak.
    - The sequence then repeats, moving from peak to recession to trough to recovery and back to peak.
- The swings happen around a rising straight line labeled **trend** or **growth**. Over the long run, output tends to increase, so each later peak can be higher than the earlier one. The oscillation is the part of output that swings above and below this trend line.
- To treat the business cycle as a harmonic oscillator, the distance of output above or below the trend plays the role of x. Output above the trend is followed by a fall back toward and below it, and output below the trend is followed by a rise back toward and above it, which is the same back-and-forth pattern produced by $\dot{x} = v$ and $\dot{v} = -x$.

Section 14: Music and Harmonic Oscillators

- **Music is built from harmonic oscillators.** Every musical note is a vibration at a particular frequency. A plucked string, the column of air inside a wind instrument, and a vibrating reed each swing back and forth around a resting state, just as x does in the x-v system.
- The **pitch** of a note corresponds to its frequency. A higher-pitched note vibrates more times per second, and a lower-pitched note vibrates fewer times per second.
- Sheet music is a written record of which vibrations to produce and when. When several notes sound together, their vibrations add, the same way the 1 Hz and 4 Hz cosines add to form the model of /i/.

Section 15: Matter and the Quantum Harmonic Oscillator

- The harmonic oscillator appears in physics at the scale of atoms as the **quantum harmonic oscillator**. In quantum physics, a vibrating system cannot vibrate with just any amount of energy. It can only take certain specific energy levels, which is what **quantized** means.
- **Matter is made up of atoms with different numbers of electrons, some vibrating at a low frequency and some at a high frequency.** In the quantum description, electrons in an atom are not tiny balls on tracks but **wave modes**, which are specific allowed patterns of vibration.
- **Magnesium (Mg)** shows two ways of picturing an atom. A neutral magnesium atom has **12 electrons**, and its nucleus contains **12 protons**. Its **ground-state configuration**, the

arrangement of electrons at lowest energy, is written $1s^2\ 2s^2\ 2p^6\ 3s^2$.

  - In the older **"solar system" picture**, also called the **Bohr-style shell** picture, electrons are drawn as particles moving on circular paths around the nucleus. Magnesium's electrons are arranged in shells of **2, 8, and 2**: 2 electrons in shell 1, 8 electrons in shell 2, and 2 electrons in shell 3. This is a historical model of how the shells are arranged.
  - In the **quantum orbital picture**, each occupied wave mode, called an **orbital**, contributes a cloud of **electron density** around the nucleus. Darker shading indicates greater density, meaning a region where an electron is more likely to be found.
  - The **total electron density** of magnesium is built by adding together the densities of all its occupied orbitals: $1s^2$, $2s^2$, $2p^6$, and $3s^2$. The superscript counts the electrons in each orbital group, and $2p^6$ includes all three filled p orbitals. The sum is a sum of **probability densities**, and the sizes and shading contrasts of the orbitals are drawn schematically rather than to exact scale.
  - The innermost $1s$ electrons form the densest cloud close to the nucleus, the $2s$ and $2p$ electrons form a middle layer, and the $3s$ electrons form the faint outermost layer.
- Atomic orbitals are **quantized wave modes**, which connects them to the idea of vibration, but **magnesium is not a harmonic-oscillator atom**. The quantum harmonic oscillator is a specific model, and real atoms like magnesium follow related but different rules. The electron configuration comes from NIST atomic data for magnesium, and the quantum model follows OpenStax *University Physics* Volume 3, sections 7.5 and 8.1.

Section 16: The x-v System in Python

- The x-v system can be computed step by step in Python, producing the same sequence of points traced through time in Section 4. The system starts with x at its maximum and v at zero, and time is divided into many small steps of size `dt`.
  - At each step, $\dot{x}$ is set equal to the current v, following the first equation.
  - At the same step, $\dot{v}$ is set equal to the negative of the current x, following the second equation.
  - The new x is the old x plus $\dot{x}$ multiplied by the step size, and the new v is the old v plus $\dot{v}$ multiplied by the step size.
  - Repeating this many times builds up the full x wave and v wave one small step at a time. Smaller steps make the computed points follow the true smooth waves more closely.

python

```python
import numpy as np
import matplotlib.pyplot as plt

dt = 0.01        # size of each time step
steps = 1000     # number of steps to compute

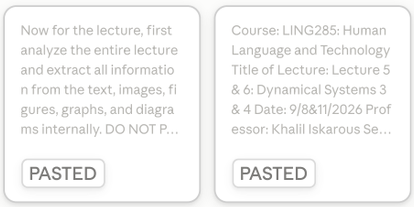
x = np.zeros(steps)
v = np.zeros(steps)
x[0] = 1.0       # x starts at its maximum
v[0] = 0.0       # v starts at zero

for n in range(steps - 1):
    x_dot = v[n]         # x-dot = v
    v_dot = -x[n]        # v-dot = -x
    x[n + 1] = x[n] + x_dot * dt
    v[n + 1] = v[n] + v_dot * dt

t = np.arange(steps) * dt
plt.plot(t, x, label="x")
plt.plot(t, v, label="v")
plt.xlabel("t")
```

```
plt.legend()
plt.show()
```

- The plotted output reproduces the behavior of the harmonic oscillator. x starts at the top and traces a cosine wave, and v starts at zero, drops negative first, and traces the same wave shape shifted by a quarter of a cycle. v reaches its extremes whenever x crosses zero, and v passes through zero whenever x reaches its extremes.



Course: LING285: Human Language and Technology

Title of Lecture: Lecture 5 & 6: Dynamical Systems 3 & 4

Date: 9/8&11/2026

Professor: Khalil Iskarous

Section 1: The Elements of the Harmonic Oscillator, x and v

- A **harmonic oscillator** is built from two quantities, x and v. x is how far something is displaced from its resting point, and v is its **velocity**, meaning how fast and in which direction x is moving. The system is about **how x and v change over time as a result of their relation to each other**.
- A dot over a variable means its change over time. $\dot{x}$ is the change in x as time increases, and $\dot{v}$ is the change in v as time increases. This is **Isaac Newton's notation**.
  - When $\dot{x}$ is positive, x rises, and when $\dot{x}$ is negative, x falls.
  - The same rule holds for v. A positive $\dot{v}$ means v rises, and a negative $\dot{v}$ means v falls.
- The defining feature of the harmonic oscillator is that x **and** v **change each other**. The rate at which x changes is set by v, and the rate at which v changes is set by x. This two-way dependence produces repeating back-and-forth motion.

Section 2: The Two Equations of the Harmonic Oscillator

- The first equation says that the rate of change of x equals the current value of v:

$$\dot{x} = v$$

- When v is positive, $\dot{x}$ is positive, so x rises.
- When v is negative, $\dot{x}$ is negative, so x falls.
- The sign of v at any moment is the direction x moves at that moment.
- The second equation says that the rate of change of v equals the negative of x:

$$\dot{v} = -x$$

- The **minus sign** means v is always pushed in the direction opposite to wherever x currently is.
- When x is positive, $\dot{v}$ is negative, so v falls. Once v becomes negative, the first equation makes x fall.
- When x is negative, $\dot{v}$ is positive, so v rises. Once v becomes positive, the first equation makes x rise.
- Together, the two equations form a **restoring relationship**. Whenever x moves away from zero in either direction, the second equation turns v around, and the first equation then carries x back toward zero. This is what makes the system swing back and forth instead of drifting away.

- The relationship forms a **closed loop of influence** between x and v. One arrow runs from v to x, standing for $\dot{x} = v$, and a second arrow runs from x back to v, standing for $\dot{v} = -x$. A change in v changes x, and that change in x goes on to change v again.

Section 3: A Worked Example with x and v

- Suppose v equals -5. Substituting into the first equation gives:

$$\dot{x} = v = -5$$

- The rate of change of x is negative, so x **will go down**. Only the sign of v decides whether x is rising or falling.
- Suppose x equals -5. Substituting into the second equation gives:

$$\dot{v} = -x = -(-5) = 5$$

- The two minus signs cancel, so the rate of change of v is positive, and v **will go UP**.
- This is the restoring behavior. When x is below zero, v is pushed upward, and a positive v then lifts x back toward zero through the first equation.

Section 4: The x-v System Traced Through Time

- The system can be followed moment by moment, starting with x at its highest positive value and v equal to zero.
  - At the start, x is positive, so $\dot{v}$ is negative and v drops below zero. With v negative, $\dot{x}$ is negative, so x begins to decrease, slowly at first.
  - While x is still positive, v keeps getting more negative, so x falls more and more steeply.
  - When x reaches zero, $\dot{v}$ equals zero, so v stops decreasing. At this instant v is at its most negative value, and x is crossing zero at its fastest speed.
  - Once x is below zero, $\dot{v}$ is positive, so v rises back toward zero. x keeps falling, but more and more slowly.
  - When v returns to zero, x is at its lowest value and turns around.
  - With x negative, v keeps rising and becomes positive, so x climbs. v reaches its largest positive value as x crosses zero on the way up, and x returns to its maximum as v comes back to zero. The cycle then repeats.
- Computing x and v at a series of separate moments gives a sequence of points, where each new value is worked out from the values just before it. These points trace two waves.
  - The x wave starts at the top, dips below zero to a lowest point, and climbs back past zero to the top again.
  - The v wave starts at zero, dips below zero to its lowest point, returns to zero, and rises above zero.
  - The two waves have the same shape but are shifted by a quarter of a cycle. v is at its extremes when x crosses zero, and v is zero when x is at its extremes.

Section 5: The Oscillation/Vibration Dynamical System

- A **dynamical system** is a set of variables together with rules for how those variables change over time. The x-v system is an **oscillation** or **vibration dynamical system**, meaning a quantity repeatedly swings back and forth around a middle value.
- The loop between x and v generates this swinging by itself. x drives changes in v, and v drives changes in x, so once the system is set in motion it keeps cycling.
- Plotted against time t, x traces a wave that starts at its peak, falls below zero, rises back to the same peak, and repeats. Each cycle reaches the same height and depth and takes the same amount of time. v follows its own repeating wave over the same stretch of time.

Section 6: Everything's a Harmonic Oscillator, If You're Brave Enough

- The two-variable structure of $\dot{x} = v$ and $\dot{v} = -x$ appears in many systems, including the air in the vocal tract, the vocal folds, economies, music, and matter.

- Treating a system as a harmonic oscillator means identifying something that plays the role of x, a displacement from a resting state, and something that plays the role of v, the rate at which that displacement changes. Real systems are usually more complicated than two linked variables, so the harmonic oscillator is a simplified model of their back-and-forth pattern.

Section 7: A Slice of Air in the Vocal Tract

- The **vocal tract** is the air-filled passage from the vocal folds in the throat up through the mouth to the lips. A thin slice of air inside it contains many air particles.
- **Pressure** in the slice depends on how close together the particles are.
  - **High pressure** means the air particles are very close together.
  - **Low pressure** means the air particles are very far apart.
  - The particles stay the same in number and size. Only the distances between them change.
- **Compression** is the state in which particles are closer together than their normal spacing. **Rarefaction** is the state in which particles are farther apart than their normal spacing. The normal spacing, where the air is neither squeezed nor stretched, is the **equilibrium spacing**.
  - A tracked pair of neighboring particles sits at 100% of its equilibrium distance at equilibrium.
  - At the strongest compression, the pair is squeezed to about 56% of its equilibrium distance.
  - At the strongest rarefaction, the pair is stretched to about 141% of its equilibrium distance.
  - The slice cycles continuously between compression and rarefaction, passing through equilibrium each time. These values are exaggerated, and the slice is an idealized piece of air.
- The slice of air behaves like a harmonic oscillator. The particle spacing relative to equilibrium plays the role of x. Squeezed air pushes itself apart, and spread-out air is pulled back together, so any departure from equilibrium is pushed back toward equilibrium. This is the restoring relationship expressed by $\dot{v} = -x$.

Section 8: There's a Cosine in Your Mouth

- The back-and-forth motion of the air slice follows a **cosine** function:

$$x = \cos(\text{phase})$$

- Here x measures the **normalized expansion** of the slice, meaning the values are scaled so the largest stretch equals $+1$ and the largest squeeze equals -1.
  - $x = +1$ means the particles are **farthest apart**, which is maximum rarefaction.
  - $x = 0$ means the particles are at **equilibrium** spacing.
  - $x = -1$ means the particles are **closest together**, which is maximum compression.
  - Positive values of x mean the slice is rarefied, and negative values mean it is compressed.
- The **phase** is the position within a cycle. One complete cycle corresponds to a phase increase of 2π, so a phase running from 0 to 6π covers three full cycles, ending at 2π, 4π, and 6π.
- One cycle proceeds in order from maximum rarefaction to compression and back to rarefaction.
  - At phase 0, the cosine equals $+1$, so the slice is at maximum rarefaction.
  - As the phase increases, x decreases. At $x = +0.74$, the particles are still farther apart than equilibrium, but less so than at the start.
  - At phase $\pi/2$, x equals 0, and the slice passes through equilibrium.
  - At $x = -0.54$, the slice is compressed.
  - At phase π, x equals -1, which is maximum compression.
  - At phase $3\pi/2$, x returns to 0, and at phase 2π, x is back at $+1$.

    - In the following cycles, values such as $+0.31$ and $+0.45$ mean rarefaction, and values such as -0.05 and -0.21 mean compression.
- The cosine is the same wave the x-v system produces when x starts at its maximum and v starts at zero. The vibrating air in the vocal tract follows the harmonic oscillator relationship, and its expansion over time traces a cosine.

Section 9: Vowels and the Shape of the Vocal Tract

- Vowels differ in where the **tongue** sits in the vocal tract. Images taken down the middle of the head from front to back capture the tongue, the roof of the mouth, and the throat during each vowel.
- Vowels are organized by **height**, whether the tongue body is raised (**high**) or lowered (**low**), and by **frontness**, whether the tongue body is forward (**front**), pulled back (**back**), or in between (**central**).
    - In **"dean,"** the vowel is **high and front**. The tongue is raised and pushed forward toward the roof of the mouth, leaving a large open space in the throat behind it.
    - In **"dune,"** the vowel is **high and back**. The tongue is raised, but its highest point is pulled back toward the rear of the roof of the mouth.
    - In **"Dan,"** the vowel is **low and front**. The tongue is lowered and the mouth is more open, with the tongue body positioned toward the front.
    - In **"Don,"** the vowel is **low and back**. The tongue is lowered and pulled back into the throat, narrowing the space behind it.
    - In **"done,"** the vowel is **central**. The tongue rests in the middle of the mouth.
- Each tongue position gives the air in the vocal tract a different shape, and a differently shaped column of air vibrates differently. This difference in vibration distinguishes one vowel from another.

Section 10: Vowels as Combined Vibrations

- A vowel is a **combination of vibrations.** The air in the vocal tract vibrates at several frequencies at once, and the vowel is the **sum** of those vibrations. Each individual vibration is a cosine, so a vowel can be modeled as harmonic oscillators added together.
- **Frequency** is the number of complete cycles per second, measured in **hertz (Hz).** A vibration at f hertz is written $\cos(2\pi f t)$, where t is time in seconds and 2π converts cycles into phase.
- The vowel /**i**/, as in "dean," is modeled as a 1 Hz vibration plus a 4 Hz vibration:

$$x(t) = \cos(2\pi t) + \cos(8\pi t)$$

- The first term is the 1 Hz component, since $2\pi \times 1 = 2\pi$, and the second term is the 4 Hz component, since $2\pi \times 4 = 8\pi$.
- The sum is a slow wave with fast ripples on top. Over 3 seconds, the slow component completes 3 cycles and the fast component completes 12 cycles.
- The vowel /**a**/, as in "Don," is modeled as a 2 Hz vibration plus a 3 Hz vibration:

$$x(t) = \cos(4\pi t) + \cos(6\pi t)$$

- The first term is the 2 Hz component, since $2\pi \times 2 = 4\pi$, and the second term is the 3 Hz component, since $2\pi \times 3 = 6\pi$.
- Over 3 seconds, the 2 Hz component completes 6 cycles and the 3 Hz component completes 9 cycles, producing a different pattern of tall and short swings than /i/.
- These are **illustrative two-frequency models** with **equal component amplitudes**, so each cosine swings between $+1$ and -1.
    - The sum therefore ranges between $+2$ and -2.
    - The sum reaches $+2$ when both cosines peak together, which happens for both vowels at $t = 0, 1, 2,$ and 3 seconds.
    - At other moments, the components partly cancel or reinforce. /i/ can be at -1.21 while /a/ is at $+0.58$, and /i/ can be at $+0.62$ while /a/ is at -1.62.
- **Positive x means rarefaction** and **negative x means compression.** /i/ and /a/ squeeze and stretch the air slice in two different time patterns, and these patterns distinguish the

two vowels.

Section 11: Praat Demonstration

- **Praat** is a computer program for analyzing speech. Recorded vowels can be examined in Praat to see their vibrations, which are the combined vibrations modeled for /i/ and /a/.

Section 12: Two Things Are Vibrating, the Vocal Tract Air and the Vocal Folds

- During speech, **two things vibrate**. One is the air in the vocal tract, which is the vibration modeled by cosines and sums of cosines. The other is the **vocal folds**, the tissue in the voice box of the throat that vibrates during speech.

Section 13: The Harmonic Oscillator in Economics

- The **business cycle** is the repeated rise and fall of an economy's **level of real output** over **time**. Real output is the amount of goods and services an economy produces.
- The business cycle moves through repeating phases.
  - At a **peak**, output is at a high point.
  - During a **recession**, output falls.
  - At a **trough**, output reaches a low point.
  - During a **recovery**, output rises to the next peak, and the sequence repeats.
- The swings occur around a rising **trend** line labeled **growth**. Output increases over the long run, so a later peak can be higher than an earlier one. The oscillation is the swinging of output above and below this trend.
- As a harmonic oscillator, the distance of output above or below the trend plays the role of x. Output above the trend is followed by a fall below it, and output below the trend is followed by a rise above it.

Section 14: Music and Harmonic Oscillators

- **Music is all about harmonic oscillators.** Each musical note written in sheet music is a vibration at a particular frequency, and a piece of music is a sequence and combination of these vibrations.

Section 15: Matter and the Quantum Harmonic Oscillator

- The harmonic oscillator also applies to matter through the **quantum harmonic oscillator**. **Matter is made up of atoms with different numbers of electrons, some vibrating at a low frequency and some at a high frequency.**
- **Magnesium (Mg)** can be pictured in two ways. A neutral magnesium atom has **12 electrons** and **12 protons**, with the ground-state configuration $1s^2\ 2s^2\ 2p^6\ 3s^2$.
  - In the **"solar system" picture**, or **Bohr-style shell** picture, electrons are particles moving on paths around the nucleus. Magnesium's shells hold **2, 8, and 2** electrons: 2 in shell 1, 8 in shell 2, and 2 in shell 3. This is a historical model of the shell arrangement.
  - In the **quantum orbital picture**, each occupied **wave mode**, called an **orbital**, contributes **electron density** around the nucleus. Darker shading means greater density.
  - The **total electron density** is the sum of the occupied orbital densities $1s^2$, $2s^2$, $2p^6$, and $3s^2$, where $2p^6$ includes all three filled p orbitals. This sum combines **probability densities**, and orbital sizes and density contrasts are schematic.
- **Atomic orbitals are quantized wave modes, but magnesium is not a harmonic-oscillator atom.** The configuration comes from NIST atomic data for magnesium, and the quantum model follows OpenStax *University Physics* Volume 3, sections 7.5 and 8.1.

Section 16: The x-v System in Python

- The x-v system can be computed in Python one moment at a time, starting with x at its maximum and v at zero.
  - At each step, x changes according to $\dot{x} = v$, and v changes according to $\dot{v} = -x$.

- Each new pair of values is computed from the pair before it, producing a sequence of points.
- The x points trace a cosine wave that starts at the top, and the v points start at zero, drop negative first, and trace the same wave shape shifted by a quarter of a cycle.